Extremally projective normed spaces

A characterization in all dimensions

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Helemskii's seminar

Plan

1. Lifting problems
2. Basic theory
3. Dimension restrictions
4. Countable dimension
5. Extremally projective spaces which are not free
6. Uncountable dimension
7. Summary

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6. Uncountable dimension

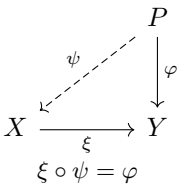
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Lifting problems

Given a "surjection" $\xi\colon X \to Y$ and a map $\varphi\colon P \to Y$: does φ lift to X?

P is projective if every such problem has a solution ψ with $\|\psi\| \le cC\|\varphi\|$.

Which surjections? In $\mathbf{Nor}$, the category of *normed* spaces and bounded operators, the right notion is quantitative.



Definition

$\xi\colon X \to Y$ is c-topologically surjective ($c \ge 1$) if $c\,\xi(B_X^\circ) \supset B_Y^\circ$, i.e.

$$\forall y \in Y \quad \forall \delta > 0 \quad \exists x \in X : \quad \xi(x) = y, \qquad \|x\| < c\|y\| + \delta.$$

Coisometric: contractive and 1-topologically surjective, $\xi(B_X^\circ) = B_Y^\circ$.

Definition

A normed space P is C-topologically projective ($C \geq 1$) if for every $c \geq 1$, every c-topologically surjective $\xi\colon X \to Y$ between normed spaces and every bounded $\varphi\colon P \to Y$ there is a bounded $\psi\colon P \to X$ with

$$\xi\psi = \varphi, \qquad \|\psi\| \leq cC\|\varphi\|.$$

P is topologically projective if this holds for some C, and extremally projective if it holds for *every* $C > 1$.

- X and Y are arbitrary *normed* spaces. Restricting to Banach spaces gives the (different!) theory in $\mathbf{Ban}$.
- Normed spaces = normed modules over the zero algebra. This is the simplest case of Helemskii's *extreme projectivity* for normed modules; the two definitions agree (later).
- Extremal projectivity sits between the metric theory ($C = 1$) and the topological theory (some C).

Banach spaces: the classical answers

Theorem (Köthe 1966)

A Banach space is topologically projective in $\mathbf{Ban}$ $\iff$ *it is topologically isomorphic to* $\ell_1(\Lambda)$ *for some set* Λ.

Theorem (Grothendieck 1955)

A Banach space has the metric lifting property $\iff$ *it is isometrically isomorphic to some* $\ell_1(\Lambda)$.

Both proofs use completeness: open mapping, Pełczyński's decomposition method.

In $\mathbf{Nor}$ the free objects are different

$$\ell_1^0(\Lambda) = \{\text{finitely supported families on } \Lambda\}, \qquad \|x\| = \sum_\lambda |x_\lambda|.$$

Bounded families $(x_\lambda)_{\lambda\in\Lambda}$ in a *normed* space $E \leftrightarrow$ bounded operators $T\colon \ell_1^0(\Lambda) \to E$, $T(e_\lambda) = x_\lambda$, $\|T\| = \sup_\lambda \|x_\lambda\|$. No completeness of E is needed, in contrast with $\ell_1(\Lambda)$.

Normed spaces: what was known

Theorem (Grønbæk 2016)

E is topologically projective in $\mathbf{Nor}$ $\iff$ *E is topologically isomorphic to* $\ell_1^0(\Lambda)$, $|\Lambda| = \operatorname{dens}(E)$.

Theorem (Helemskii 2013)

E is metrically projective in $\mathbf{Nor}$ $\iff$ *E is isometrically isomorphic to some* $\ell_1^0(\Lambda)$.

Extremal projectivity (Helemskii 2013): $\ell_1^0(\Lambda)$ is extremally projective; P extremally projective $\iff$ P is a near-retract of some $\ell_1^0(\Lambda)$; P extremally projective $\Rightarrow \overline{P} \cong \ell_1(\Lambda)$ isometrically, so P is a dense subspace of some $\ell_1(\Lambda)$; P separable and topologically projective $\Rightarrow \dim P \leq \aleph_0$.

Question (Helemskii 2013; Grønbæk 2016, Conjecture 3.1)

Is every extremally projective normed space isometrically isomorphic to some $\ell_1^0(\Lambda)$? More precisely: *which dense subspaces of* $\ell_1(\Lambda)$ *are extremally projective?*

The answers, I

Theorem A (counterexample)

$\ell_1^0(\mathbb{N}) + \mathbb{C}v$, $v = (2^{-n})_n$, is extremally projective but not isometrically isomorphic to any $\ell_1^0(\Lambda)$.

Theorem B (almost isometric characterization)

For a normed space P the following are equivalent:

(i) P is extremally projective;

(ii) P is topologically projective and $\overline{P} \cong \ell_1(\Gamma)$ isometrically;

(iii) $d_{BM}(P, \ell_1^0(\Gamma)) = 1$ for some Γ;

(iv) for every $\varepsilon > 0$ there is Λ with $d_{BM}(P, \ell_1^0(\Lambda)) < 1 + \varepsilon$.

For $\dim P \le \aleph_0$: P extremally projective $\iff$ $\overline{P} \cong \ell_1$ or ℓ_1^n isometrically.

So Grønbæk's conjecture fails isometrically and holds almost isometrically.

The answers, II

Theorem C (dimension restrictions)

Topologically projective normed spaces contain no infinite dimensional complete subspaces. In particular, no infinite dimensional Banach space is topologically projective in $\mathbf{Nor}$.

Theorem D (the uncountable case is different)

There is a dense subspace of $\ell_1(\omega_1)$, all of whose separable subspaces have countable dimension, which is not topologically projective.

Theorem B says that for dense subspaces of $\ell_1(\Gamma)$ topological and extremal projectivity coincide. Theorem D says that for uncountable Γ topological projectivity is a genuine restriction beyond the dimension bound of Theorem C.

Why "normed" is the difficulty

Theorem C says: an extremally projective space never contains an infinite dimensional Banach space. So the objects are *incomplete in every direction*, and the usual toolbox is switched off:

- no Baire category, no open mapping, no closed graph;
- no Pełczyński decomposition method (ℓ_1-sums of the pieces are not inside P);
- no Grothendieck-type duality for P itself, only for $\overline{P}$.

What survives

The finite dimensional pieces of P, the Hamel basis, Hahn–Banach, the Neumann series in the *completion*, and the exact ℓ_1-norm.

The main tool will be a back-and-forth construction: an automorphism of $\ell_1(\Gamma)$ of distortion $< 1+\varepsilon$ built step by step, which moves P exactly onto $\ell_1^0(\Gamma)$. In the uncountable case the steps are indexed by ordinals and controlled on disjoint blocks of coordinates.

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Helemskii's definition is the same

Helemskii: P is *extremely projective* if for every *coisometric* $\xi\colon X \to Y$, every $\varphi\colon P \to Y$ and every $\delta > 0$ there is a lifting ψ with $\|\psi\| < \|\varphi\| + \delta$.

Proposition

Extremally projective $\iff$ *extremely projective in the sense of Helemskii.*

Proof sketch.

($\Rightarrow$) Coisometric operators are 1-topologically surjective; take $C = 1 + \delta/(2\|\varphi\|)$.
($\Leftarrow$) Given a c-topologically surjective ξ, renorm X:

$$\|x\|' = \max\bigl(\|x\|,\ c\|\xi(x)\|\bigr).$$

Then $\xi' = c\xi\colon X' \to Y$ is coisometric: contractive by construction, and $y \in B_Y^\circ$ has a preimage $x \in B_X^\circ$ under $c\xi$ with $\|x\|' = \max(\|x\|, \|y\|) < 1$. A lifting ψ' along ξ' with $\|\psi'\|' < C\|\varphi\|$ gives $\psi = c\psi'$ with $\xi\psi = \varphi$ and $\|\psi\| \le c\|\psi'\|' \le cC\|\varphi\|$. □

The renorming trick will reappear: it makes the two conventions (open ball, closed ball) interchangeable for extremal projectivity.

Free spaces are extremally projective

Proposition

$\ell_1^0(\Lambda)$ is $(1+\varepsilon)$-topologically projective for every $\varepsilon > 0$.

Proof.

Let $\xi\colon X \to Y$ be c-topologically surjective and $\varphi\colon \ell_1^0(\Lambda) \to Y$, $\varphi \neq 0$. For each λ choose $x_\lambda \in X$ with

$$\xi(x_\lambda) = \varphi(e_\lambda), \qquad \|x_\lambda\| < c\|\varphi(e_\lambda)\| + c\varepsilon\|\varphi\| \leq c(1+\varepsilon)\|\varphi\|.$$

The family is bounded, so $\psi(e_\lambda) := x_\lambda$ defines a bounded operator with $\|\psi\| = \sup_\lambda \|x_\lambda\| \leq c(1+\varepsilon)\|\varphi\|$ and $\xi\psi = \varphi$ on the Hamel basis. □

The constant cannot be 1

The only 1-topologically projective normed space is $\{0\}$: a coisometric functional on $(c_{00}, \|\cdot\|_\infty)$ need not attain its norm (backup slide). So in $\mathbf{Nor}$ one considers all $C > 1$ at once.

Retracts and the canonical morphism

$\zeta\colon E \to F$ is a c-retraction if $\zeta\eta = 1_F$ for some η with $\|\zeta\|\|\eta\| \le c$; then F is a c-*retract* of E.

Proposition (permanence)

P' C-topologically projective, P a c-retract of $P' \Rightarrow P$ is cC-topologically projective.
Hence: a $(1+\varepsilon)$-retract of an extremally projective space for every $\varepsilon > 0$ is extremally projective.

For a normed space P consider the canonical morphism

$$\pi_P\colon \ell_1^0(B_P) \to P, \qquad e_x \mapsto x.$$

Contractive, and $\pi_P(B_{\ell_1^0(B_P)}) = B_P$: strictly coisometric.

Proposition

(i) *P C-topologically projective $\Rightarrow$ π_P has a right inverse σ with $\|\sigma\| \le C$ (lift 1_P along π_P);*
(ii) *π_P has a right inverse of norm $\le C \Rightarrow P$ is $C(1+\varepsilon)$-topologically projective for all $\varepsilon > 0$.*

One test case suffices: the universal surjection π_P.

The criterion

Theorem

For a normed space P the following are equivalent:

(i) *P is extremally projective;*

(ii) *for every $\varepsilon > 0$, π_P has a right inverse of norm $\leq 1+\varepsilon$;*

(iii) *for every $\varepsilon > 0$, P is a $(1+\varepsilon)$-retract of some $\ell_1^0(\Lambda)$;*

(iv) *for every $\varepsilon > 0$ there are Λ, a subspace $Z \subset \ell_1^0(\Lambda)$ and a projection Q of $\ell_1^0(\Lambda)$ onto Z with $\|Q\| \leq 1+\varepsilon$ and $d_{BM}(P, Z) \leq 1+\varepsilon$.*

$(i) \Leftrightarrow (iii)$ is Helemskii's near-retract criterion. $(iii) \Rightarrow (iv)$: normalize $\|\zeta\| = 1$, $\|\eta\| \leq 1+\varepsilon$, put $Z = \eta(P)$, $Q = \eta\zeta$. $(iv) \Rightarrow (i)$: $\zeta = T^{-1}Q$, $\eta = T$ make P a $(1+\varepsilon)^2$-retract of $\ell_1^0(\Lambda)$; use permanence.

Takeaway

For every ε, P sits inside a free space $(1+\varepsilon)$-isomorphically and $(1+\varepsilon)$-complemented. We shall construct **isomorphisms onto free spaces** of distortion $< 1+\varepsilon$.

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Free spaces are small

For infinite Λ: $\dim \ell_1^0(\Lambda) = |\Lambda| = \operatorname{dens} \ell_1^0(\Lambda)$. An infinite dimensional Banach space has uncountable Hamel dimension (Baire), in fact $\geq \mathfrak{c}$ (Löwig 1934).

Theorem (hereditary dimension bound)

Let P be topologically projective and $Z \subset P$ a subspace. Then

$$\dim Z \leq \max(\operatorname{dens} Z, \aleph_0).$$

In particular every separable subspace of P has countable dimension.

Proof.

Take $\sigma\colon P \to \ell_1^0(B_P)$ with $\pi_P \sigma = 1_P$; σ is injective. Let $M \subset Z$ be dense, $|M| = \operatorname{dens} Z$, and $S_0 = \bigcup_{x \in M} \operatorname{supp} \sigma(x)$, so $|S_0| \leq \max(\operatorname{dens} Z, \aleph_0)$. For $z \in Z$, $\sigma(z)$ is a limit of elements supported in S_0; coordinate functionals are continuous, so $\sigma(z)$ is supported in S_0. Hence $\sigma(Z) \subset \ell_1^0(S_0)$, a space of dimension $|S_0|$. □

Helemskii's argument for separable P, in the hereditary form needed later.

No complete pieces

Corollary

Let P be topologically projective, $Z \subset P$, and $\|\cdot\|'$ a norm on Z with $(Z, \|\cdot\|')$ complete and $\|\cdot\| \leq K\|\cdot\|'$. Then $\dim Z < \infty$.

Proof.

A separable infinite dimensional $\|\cdot\|'$-closed subspace Z_0 would be $\|\cdot\|$-separable with uncountable dimension. □

Every complete subspace of P is finite dimensional. No infinite dimensional Banach space is topologically projective in **Nor**, although $\ell_1(\Lambda)$ is extremally projective in **Ban**.

Example: completions do not decide

$w_n \geq 1$, $w_n \to \infty$, $P_w = \{x \in \ell_1 : \sum_n w_n|x_n| < \infty\}$ with the ℓ_1-norm: dense in ℓ_1, contains $\ell_1^0(\mathbb{N})$, so $\overline{P_w} = \ell_1$; every complete subspace is finite dimensional (open mapping in $\ell_1(w)$); not topologically projective, as P_w is separable but $\dim P_w = \dim \ell_1(w) > \aleph_0$.

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Theorem (Helemskii 2013; Grønbæk 2016)

P extremally projective $\Rightarrow \overline{P} \cong \ell_1(\Gamma)$ *isometrically for some set* Γ.

Sketch. By the criterion, $\overline{P}$ is a $(1+\varepsilon)$-retract of the Banach space $\ell_1(B_P)$ for every ε. Then

$$\overline{P}^* \text{ is } (1+\varepsilon)\text{-injective } \forall\varepsilon \;\Rightarrow\; 1\text{-injective} \;\Rightarrow\; \overline{P}^* \cong C(K),\ K \text{ Stonean}$$
$$\Rightarrow\; \overline{P} \cong L_1(\mu) \;\Rightarrow\; \mu \text{ atomic},$$

the last step because $\overline{P}$ has the Schur property as a complemented subspace of $\ell_1(B_P)$ (Grothendieck; Lacey, Thms 3.11.6 and §17).

Two sides of the picture

An extremally projective normed space is, up to isometry,

a **dense subspace of** $\ell_1(\Gamma)$ whose **separable subspaces have countable dimension**.

Question. Is every such subspace extremally projective?
Countable Γ: ✓, and more. Uncountable Γ: ×, but a correction works.

Theorem (back-and-forth)

Let $D \subset \ell_1$ be a dense subspace of countable dimension and $\varepsilon > 0$. There is a linear bijection T of ℓ_1 with

$$\|T\|\|T^{-1}\| \le 1+\varepsilon, \qquad T(D) = \ell_1^0(\mathbb{N}).$$

In particular $d_{BM}(D, \ell_1^0(\mathbb{N})) = 1$.

Plan. (d_j) a Hamel basis of D; $\theta_m \in (0, 1/2]$ with $\prod_m (1+\theta_m)/(1-\theta_m) \le 1+\varepsilon$. Build automorphisms $F_1, F_2, \dots$ of ℓ_1 with $\|F_m - 1\| \le \theta_m$ and put $A_j = F_{2j}F_{2j-1}A_{j-1}$, $T = \lim A_j$.

forth (F_{2j-1}): make $A_{j-1}(d_j)$ finitely supported;

back (F_{2j}): make e_j an element of $A_j(D)$;

while **fixing** a growing finite dimensional $Y_{j-1} \subset \ell_1^0(\mathbb{N})$ that records what was already achieved.

Three lemmas: a *one-vector* lemma, a *hitting* lemma, and a *product* lemma.

The one-vector lemma

Notation: $x|_N = x$ truncated after coordinate N.

Lemma

Let $v \in \ell_1$, $N \geq 0$, $0 < \delta < 1$. There is $R\colon \ell_1 \to \ell_1$, $\|R\| \leq \delta$, with

(i) $R(x) = 0$ *whenever* $x = x|_N$;

(ii) $R(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N})$;

(iii) $(1 - R)(v) \in \ell_1^0(\mathbb{N})$;

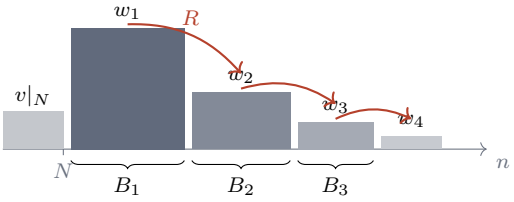
(iv) $(1 - R)^{-1}(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N}) + \mathbb{C}v$.

Hence $T = 1 - R$ is an automorphism of ℓ_1, $\|T\| \leq 1 + \delta$, $\|T^{-1}\| \leq (1 - \delta)^{-1}$, T fixes everything supported in $[1, N]$, $T(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N})$ and

$$T(\ell_1^0(\mathbb{N}) + \mathbb{C}v) = \ell_1^0(\mathbb{N}).$$

One small perturbation of the identity swallows one vector into $\ell_1^0(\mathbb{N})$, without disturbing a prescribed finite block of coordinates and without letting anything else out of $\ell_1^0(\mathbb{N})$ except multiples of v.

Proof: shift along blocks



Cut the tail $v - v|_N$ into finite blocks $w_k = v|_{B_k}$ with

$$\|w_{k+1}\| \le \delta \|w_k\|.$$

Possible since $t_M = \sum_{n>M} |v_n| \downarrow 0$.

$g_k(x) = \sum_{n \in B_k} \overline{\operatorname{sgn} v_n}\, x_n$: then
$g_k(w_j) = \delta_{jk}\|w_k\|$, $|g_k(x)| \le \sum_{B_k} |x_n|$.

$$R(x) = \sum_{k \ge 1} \frac{g_k(x)}{\|w_k\|}\, w_{k+1}, \qquad \|R\| \le \delta.$$

- $R(w_k) = w_{k+1}$: R shifts the blocks one step to the right.
- $R(v) = \sum_k w_{k+1} = v - v|_N - w_1$, so $(1-R)(v) = v|_N + w_1 \in \ell_1^0(\mathbb{N})$.
- For $x \in \ell_1^0(\mathbb{N})$: $R^m(x) = \sum_{k \in K} c_k w_{k+m}$ (finite K), so $(1-R)^{-1}x = \sum_m R^m x = x + \sum_{k \in K} c_k (v - v|_{N_k}) \in \ell_1^0(\mathbb{N}) + \mathbb{C}v$.

Lemma (hitting)

Let $Y \subset \ell_1^0(\mathbb{N})$ be finite dimensional and $e_n \notin Y$. There is $C \geq 1$ such that for every $z \in \ell_1^0(\mathbb{N})$ with $\|z - e_n\| \leq (4C)^{-1}$ there is an automorphism S of ℓ_1 with

$$\|S - 1\| \leq 2C\|z - e_n\| \leq \tfrac{1}{2}, \qquad S|_Y = 1_Y, \qquad S(z) = e_n.$$

Proof. Hahn–Banach: $f \in \ell_1^*$, $f|_Y = 0$, $f(e_n) = 1$, $\|f\| = d^{-1}$, $d = \operatorname{dist}(e_n, Y)$; $C = \max(1, d^{-1})$. Then $|f(z)| \geq 3/4$ and

$$S(x) = x + \frac{f(x)}{f(z)}\,(e_n - z)$$

does the job; it is invertible by the Neumann series. □

A finite dimensional obstacle Y is avoided by a rank-one correction; the closer z is to e_n, the smaller the correction. Density of D will supply z.

The product lemma

Lemma (product)

X Banach, $\theta_m = \|F_m - 1_X\| < 1$, $\sum_m \theta_m < \infty$, $A_j = F_j \cdots F_1$. Then $A_j \to T$ in norm, T is an automorphism, and

$$\|T\| \le \prod_m (1+\theta_m), \qquad \|T^{-1}\| \le \prod_m (1-\theta_m)^{-1}.$$

Completeness is used here, in $\ell_1 = \overline{D}$, never in D.

Choose the budgets $\theta_m \in (0, 1/2]$ in advance with $\prod_m (1+\theta_m)/(1-\theta_m) \le 1+\varepsilon$. Each step may spend at most θ_m on its perturbation; the one-vector lemma ($\|R\| \le \delta$) and the hitting lemma ($\|S-1\| \le 2C\|z - e_n\|$) respect any budget.

The back-and-forth step

Invariants for A_j and the finite dimensional $Y_j \subset \ell_1^0(\mathbb{N})$:

(a) $A_j(d_i) \in Y_j$ for $i \le j$; (forth: basis vectors are caught)

(b) $e_i \in Y_j$ for $i \le j$; (back: unit vectors are caught)

(c) $Y_j \subset A_j(D)$; (everything caught comes from D)

(d) F_{2j-1}, F_{2j} fix Y_{j-1}. (nothing caught is lost)

Forth. $N \ge$ all supports of Y_{j-1}. One-vector lemma for $v = A_{j-1}(d_j)$, N, $\delta = \theta_{2j-1}$: $F_{2j-1} = 1 - R$ fixes Y_{j-1}, and $y_j = F_{2j-1}A_{j-1}(d_j) \in \ell_1^0(\mathbb{N})$. Put $A' = F_{2j-1}A_{j-1}$, $Y' = Y_{j-1} + \mathbb{C}y_j \subset A'(D)$.

Back. If $e_j \in Y'$, done. Otherwise $A'(D)$ is dense, so pick $d \in D$ with $\|A'(d) - e_j\| \le \theta_{2j}/(16C)$.

- one-vector lemma for $A'(d)$ (fixing Y' and e_j), $\delta = \theta_{2j}/2$: $G = 1 - R'$ with $z = GA'(d) \in \ell_1^0(\mathbb{N})$, $\|z - e_j\| \le (4C)^{-1}$;
- hitting lemma: S fixes Y', $S(z) = e_j$, $\|S - 1\| \le 5\theta_{2j}/32$.

$F_{2j} = SG$, $\|F_{2j} - 1\| \le \theta_{2j}$, $Y_j = Y' + \mathbb{C}e_j$, and $e_j = A_j(d) \in A_j(D)$.

Passing to the limit

By the product lemma $A_j \to T$, an automorphism of ℓ_1 with

$$\|T\|\|T^{-1}\| \le \prod_m \frac{1+\theta_m}{1-\theta_m} \le 1+\varepsilon.$$

- $T(D) \subset \ell_1^0(\mathbb{N})$: $A_i(d_i) \in Y_i$ by (a), and all later F_m fix Y_i by (d), so $T(d_i) = A_i(d_i) \in \ell_1^0(\mathbb{N})$.
- $\ell_1^0(\mathbb{N}) \subset T(D)$: by (b), (c) there is $d \in D$ with $A_i(d) = e_i \in Y_i$, and again $T(d) = A_i(d) = e_i$.

So $T(D) = \ell_1^0(\mathbb{N})$. □

What made it work

- each step is a perturbation of the identity of size θ_m, and $\sum \theta_m$ is as small as we like: the constants are *kept*;
- each step is *local*: it fixes a finite block of coordinates and acts on the tail; the one-vector lemma produces nothing outside $\ell_1^0(\mathbb{N}) + \mathbb{C}v$;
- the Hamel basis is countable, so a single ω-sequence of steps exhausts D.

Countable dimension: the characterization

Theorem

Let $\dim P \leq \aleph_0$. *The following are equivalent:*

(i) P *is extremally projective;*

(ii) $\overline{P} \cong \ell_1(\Gamma)$ *isometrically,* Γ *countable;*

(iii) $d_{BM}(P, \ell_1^0(\Gamma)) = 1$ *for some countable* Γ*;*

(iv) *for every* $\varepsilon > 0$ *there is* Λ *with* $d_{BM}(P, \ell_1^0(\Lambda)) < 1 + \varepsilon$.

Proof. (i)⇒(ii): completion theorem; $\overline{P}$ is separable, so Γ is countable. (ii)⇒(iii): $J(P) \subset \ell_1$ is dense of countable dimension; back-and-forth. (iv)⇒(i): a $(1+\varepsilon)$-isomorphism onto $\ell_1^0(\Lambda)$ is a $(1+\varepsilon)$-retraction; criterion. □

Corollary

Every dense subspace of ℓ_1 *of countable dimension is extremally projective and has Banach–Mazur distance* 1 *to* $\ell_1^0(\mathbb{N})$.

No projectivity assumption in (ii): extremal projectivity *is* a property of the completion.

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Extreme points

By the corollary, $\ell_1^0(\mathbb{N}) + V$ is extremally projective for every $V \subset \ell_1$ of countable dimension. Is it ever free?

Lemma

Let $\ell_1^0(S) \subset E \subset \ell_1(S)$ with the ℓ_1-norm. Then

$$\operatorname{ext}(B_E) = \{\lambda e_s : \ s \in S,\ |\lambda| = 1\}, \qquad \operatorname{conv}(\operatorname{ext} B_E) = B_{\ell_1^0(S)}.$$

Proof idea.

The λe_s are extreme in $B_{\ell_1(S)}$, hence in $B_E = B_{\ell_1(S)} \cap E$. If $x \in E$, $\|x\| = 1$, has two non-zero coordinates x_n, x_m, put $t = \min(|x_n|, |x_m|)$ and

$$u = t\big(\operatorname{sgn}(x_n)e_n - \operatorname{sgn}(x_m)e_m\big) \in \ell_1^0(S) \subset E.$$

Then $\|x \pm u\| = (|x_n| \pm t) + (|x_m| \mp t) + \cdots = 1$, so x is not extreme. Finite convex combinations of the λe_s are exactly the elements of $B_{\ell_1^0(S)}$. □

The lemma only sees finitely supported vectors: E may be any space between $\ell_1^0(S)$ and $\ell_1(S)$.

The counterexample

Theorem

Let $\ell_1^0(S) \subset E \subset \ell_1(S)$ with the ℓ_1-norm. If $E \cong \ell_1^0(\Lambda)$ isometrically, then $E = \ell_1^0(S)$.

Proof.

An isometry $T\colon \ell_1^0(\Lambda) \to E$ maps extreme points onto extreme points, so

$$B_E = T\big(\operatorname{conv}\operatorname{ext} B_{\ell_1^0(\Lambda)}\big) = \operatorname{conv}\big(T \operatorname{ext} B_{\ell_1^0(\Lambda)}\big) = \operatorname{conv}\operatorname{ext} B_E = B_{\ell_1^0(S)}.$$

□

Corollary (answer to Helemskii and Grønbæk)

For $V \subset \ell_1$ of countable dimension with $V \not\subset \ell_1^0(\mathbb{N})$, e.g. $V = \mathbb{C}v$, $v = (2^{-n})$, the space $\ell_1^0(\mathbb{N}) + V$ is extremally projective and **not** isometrically isomorphic to any free normed space.

Between $\ell_1^0(\mathbb{N})$ and ℓ_1

space	extremally projective	isometric to a free space
$\ell_1^0(\mathbb{N})$	✓	✓
$\ell_1^0(\mathbb{N}) + V$, $\dim V \leq \aleph_0$, $V \not\subset \ell_1^0(\mathbb{N})$	✓	×
P_w (weighted)	×	×

All three have completion ℓ_1: the difference is invisible on completions. By the theorem, *no* space strictly between $\ell_1^0(\mathbb{N})$ and ℓ_1 is free, yet by the corollary all those of countable dimension are extremally projective.

The next section gives a criterion which distinguishes the projective ones among all dense subspaces, in any dimension.

Plan

1. Lifting problems
2. Basic theory
3. Dimension restrictions
4. Countable dimension
5. Extremally projective spaces which are not free
6. Uncountable dimension
7. Summary

What changes

Countable dimension: dense in ℓ_1 + countable dimension $\Rightarrow$ extremally projective.

Hope. Dense in $\ell_1(\Gamma)$ + all separable subspaces of countable dimension $\Rightarrow$ extremally projective?

No. For $\Gamma = \omega_1$ there is a set-theoretic refinement of the dimension bound.

Set theory in one slide

- ω_1 = the first uncountable ordinal; its elements are the countable ordinals.
- $C \subset \omega_1$ is closed unbounded (club) if unbounded and closed under countable suprema. The limit ordinals form a club.
- $E \subset \omega_1$ is stationary if it meets every club.
- **Fodor's lemma.** If E is stationary and $\gamma\colon E \to \omega_1$ is *regressive*, $\gamma(\alpha) < \alpha$, then γ is constant on a stationary subset of E.

Fodor's lemma is the uncountable analogue of the pigeonhole principle: a regressive choice cannot keep choosing new values.

The Fodor obstruction

Theorem

Let P be topologically projective, $(X_\beta)_{\beta<\omega_1}$ countable subsets of P, E a stationary set of limit ordinals, and $v_\alpha \in \overline{\operatorname{span}}\big(\bigcup_{\beta<\alpha} X_\beta\big)$ for $\alpha \in E$. Then $\operatorname{span}\{v_\alpha : \alpha \in E'\}$ has countable dimension for some stationary $E' \subset E$. In particular $(v_\alpha)_{\alpha\in E}$ is linearly dependent.

Proof.

$\sigma\colon P \to \ell_1^0(B_P)$ injective with $\pi_P\sigma = 1_P$. Put $S_\alpha = \bigcup_{\beta<\alpha}\bigcup_{x\in X_\beta} \operatorname{supp}\sigma(x)$, countable. As on the dimension slide, $\operatorname{supp}\sigma(v_\alpha) \subset S_\alpha$. But $\operatorname{supp}\sigma(v_\alpha)$ is *finite* and $S_\alpha = \bigcup_{\beta<\alpha} S_{\beta+1}$ with α a limit, so

$$\operatorname{supp}\sigma(v_\alpha) \subset S_{\gamma(\alpha)} \quad \text{for some } \gamma(\alpha) < \alpha.$$

Fodor: $\gamma \equiv \gamma_0$ on a stationary E'. Then $\sigma(v_\alpha) \in \ell_1^0(S_{\gamma_0})$ for all $\alpha \in E'$, a space of countable dimension, and σ is injective. □

Corollary

P topologically projective, $(X_\beta)_{\beta<\omega_1}$ countable subsets, $D_\alpha = \overline{\operatorname{span}}\big(\bigcup_{\beta<\alpha} X_\beta\big)$. Then

$$\big\{\alpha \text{ limit}: \ D_\alpha \neq \bigcup_{\alpha'<\alpha} D_{\alpha'}\big\} \quad \text{is not stationary.}$$

Proof.

Suppose the set E is stationary and choose $v_\alpha \in D_\alpha \setminus \bigcup_{\alpha'<\alpha} D_{\alpha'}$ for $\alpha \in E$. The family is linearly independent: in a relation $\sum_F c_\alpha v_\alpha = 0$ with $\alpha_0 = \max F$, the vector v_{α_0} would lie in D_{α_1}, $\alpha_1 = \max(F \setminus \{\alpha_0\}) < \alpha_0$, since the D_α increase. This contradicts the Fodor obstruction. □

Reading

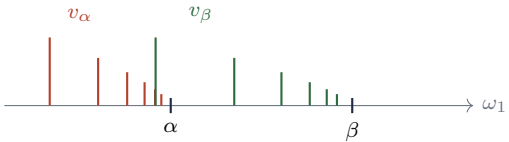
The dimension bound says: the closed span of countably many vectors of a projective space has countable dimension. The corollary adds: the closed spans of the initial segments of an ω_1-sequence are **continuous at almost all limit stages**.

The ladder space

For every limit $\alpha < \omega_1$ fix a ladder $\alpha_1 < \alpha_2 < \cdots \nearrow \alpha$ and put

$$v_\alpha = \sum_{n \geq 1} 2^{-n} e_{\alpha_n} \in \ell_1(\omega_1).$$

$$P_L = \ell_1^0(\omega_1) + \operatorname{span}\{v_\alpha : \alpha \text{ limit}\}.$$



Two ladders share only finitely many rungs: $\operatorname{supp} v_\alpha \cap \operatorname{supp} v_\beta$ is finite for $\alpha < \beta$.

(i) P_L is dense in $\ell_1(\omega_1)$.

(ii) For every countable δ: $P_L \cap \ell_1(\delta) = \ell_1^0(\delta) + \operatorname{span}\{v_\alpha : \alpha \leq \delta\}$ has countable dimension, hence is **extremally projective**; every separable subspace of P_L has countable dimension.

(iii) P_L is not topologically projective: $v_\alpha \in \overline{\operatorname{span}}\{e_\beta : \beta < \alpha\}$ for every limit α (a club), the v_α are linearly independent (almost disjoint supports), contradicting the Fodor obstruction with $X_\beta = \{e_\beta\}$.

The moral of the ladder

- P_L is "locally" extremally projective: every countably generated piece $P_L \cap \ell_1(\delta)$ is. Extremal projectivity is *not* determined by the separable subspaces.
- The chain $\delta \mapsto P_L \cap \ell_1(\delta)$ is discontinuous at *every* limit ordinal: v_δ enters at stage δ but lies in the closure of the earlier stages.
- For topologically projective spaces such discontinuities happen only on a non-stationary set (ω_1-continuity).

Strategy for the positive result

Show that for topologically projective $P \subset \ell_1(\Gamma)$ a **continuous** filtration can always be found, and run the back-and-forth construction **transfinitely** along it, with the perturbations confined to disjoint blocks of coordinates so that their "product" stays small whatever the length of the construction.

Filtrations and blocks

A filtration of Γ: an increasing family $(\Gamma_\delta)_{\delta\le\lambda}$, $\Gamma_0=\varnothing$, $\Gamma_\lambda=\Gamma$, continuous at limits. Blocks $N_\delta=\Gamma_{\delta+1}\setminus\Gamma_\delta$, so $\Gamma_\delta=\bigsqcup_{\delta'<\delta}N_{\delta'}$ and $\ell_1(\Gamma_\delta)=\big(\bigoplus_{\delta'<\delta}\ell_1(N_{\delta'})\big)_{\ell_1}$.

For $P\subset\ell_1(\Gamma)$: $P_\delta=P\cap\ell_1(\Gamma_\delta)$ and

$$D_\delta=\{x|_{N_\delta}:x\in P_{\delta+1}\}\subset\ell_1(N_\delta),\qquad D_\delta\cong P_{\delta+1}/P_\delta.$$

Theorem (transfinite back-and-forth)

Let P be dense in $\ell_1(\Gamma)$, (Γ_δ) a filtration, $0<\theta<1/2$. Suppose

(i) *P_δ is dense in $\ell_1(\Gamma_\delta)$ for all $\delta\le\lambda$;*

(ii) *$P_\delta=\bigcup_{\delta'<\delta}P_{\delta'}$ for limit $\delta\le\lambda$* *(continuity);*

(iii) *for each $\delta<\lambda$ there is an automorphism U_δ of $\ell_1(N_\delta)$ with $U_\delta(D_\delta)=\ell_1^0(N_\delta)$, $\|U_\delta\|\le1+\theta$, $\|U_\delta^{-1}\|\le1$* *(blocks are solved).*

Then there is an automorphism T of $\ell_1(\Gamma)$ with $T(P)=\ell_1^0(\Gamma)$, $\|T\|\le(1+\theta)^2$, $\|T^{-1}\|\le(1-\theta)^{-1}$.

Countable blocks: (iii) is automatic. The ladder space satisfies (i), (iii) and fails (ii).

The shape of T

Recursively choose $V_\delta\colon \ell_1(N_\delta) \to \ell_1(\Gamma_\delta)$ with $\|V_\delta\| \le \theta$, and put for $x \in \ell_1(\Gamma_\delta)$

$$W_\delta(x) = x + \sum_{\delta'<\delta} V_{\delta'}(x|_{N_{\delta'}}), \qquad U_{<\delta} = \bigoplus_{\delta'<\delta} U_{\delta'}, \qquad T_\delta = U_{<\delta} W_\delta.$$

- The $N_{\delta'}$ are disjoint and $\|x\| = \sum_{\delta'} \|x|_{N_{\delta'}}\|$, so $\|W_\delta - 1\| \le \theta$ regardless of δ: no product of λ factors ever appears.
- $\|U_{<\delta}\| \le 1+\theta$, $\|U_{<\delta}^{-1}\| \le 1$ (ℓ_1-sum).
- T_δ extends $T_{\delta'}$ for $\delta' < \delta$.

Inductive claim

$$T_\delta(P_\delta) = \ell_1^0(\Gamma_\delta) \qquad \text{for all } \delta \le \lambda.$$

Limit stages are free: by (ii) and compatibility, $T_\mu(P_\mu) = \bigcup_{\delta<\mu} T_\delta(P_\delta) = \bigcup_{\delta<\mu} \ell_1^0(\Gamma_\delta) = \ell_1^0(\Gamma_\mu)$.

Compare with the countable proof: there T was an infinite product of perturbations and the limit stage needed the product lemma. Here the perturbations live on disjoint blocks and *add* instead of multiplying.

The successor step

Let $U = U_\delta$, $y_n = U^{-1}(e_n)$ for $n \in N_\delta$: a Hamel basis of D_δ with $\|\sum c_n y_n\| \geq (1+\theta)^{-1} \sum |c_n|$.

- Lift: $x_n \in P_{\delta+1}$ with $x_n|_{N_\delta} = y_n$.
- Correct: P_δ is dense in $\ell_1(\Gamma_\delta)$ by (i), so there is $z_n \in \ell_1(\Gamma_\delta)$ with

$$x_n|_{\Gamma_\delta} + z_n \in P_\delta, \qquad \|z_n\| \leq \theta(1+\theta)^{-2}.$$

- Define $V_\delta(\sum c_n y_n) = \sum c_n W_\delta(z_n)$ on D_δ; then $\|V_\delta\| \leq \theta$, extend to $\ell_1(N_\delta)$.

Key inclusion

$$x|_{\Gamma_\delta} + W_\delta^{-1} V_\delta(x|_{N_\delta}) \in P_\delta \qquad \text{for every } x \in P_{\delta+1}.$$

($x|_{N_\delta} = \sum c_n y_n$; $x - \sum c_n x_n$ vanishes on N_δ, hence lies in P_δ.)

Then for $x \in P_{\delta+1}$:

$$T_{\delta+1}(x) = T_\delta\big(x|_{\Gamma_\delta} + W_\delta^{-1} V_\delta(x|_{N_\delta})\big) + U(x|_{N_\delta}) \in \ell_1^0(\Gamma_\delta) + \ell_1^0(N_\delta),$$

and $T_{\delta+1}(x_n) \in \ell_1^0(\Gamma_\delta) + e_n$ gives the reverse inclusion. □

Where do good filtrations come from?

Fix a dense topologically projective $P \subset \ell_1(\Gamma)$, a retraction $\sigma\colon P \to \ell_1^0(S)$, $\pi\colon \ell_1^0(S) \to P$, $\pi\sigma = 1_P$, $\|\pi\| \le 1$, and sequences $p_n^\gamma \in P$ with $p_n^\gamma \to e_\gamma$. $P_A = P \cap \ell_1(A)$.

Definition

(A, B), $A \subset \Gamma$, $B \subset S$, is closed if

(a) $\operatorname{supp} \pi(e_s) \subset A$ for $s \in B$;

(b) $\operatorname{supp} \sigma(x) \subset B$ for $x \in P_A$;

(c) $\operatorname{supp} p_n^\gamma \subset A$ for $\gamma \in A$, $n \in \mathbb{N}$.

Lemma

For a closed pair: $P_A = \pi(\ell_1^0(B))$, P_A is dense in $\ell_1(A)$, and $\sigma|_{P_A}$, $\pi|_{\ell_1^0(B)}$ exhibit P_A as a retract of $\ell_1^0(B)$. So P_A is topologically projective.

Proof. (a): $\pi(\ell_1^0(B)) \subset P_A$. (b): $x = \pi\sigma(x) \in \pi(\ell_1^0(B))$. (c): $\overline{P_A} \ni e_\gamma$ for $\gamma \in A$. □

Closed pairs are the "elementary submodels" of the situation.

Closing off, and continuity for free

Lemma (closing off)

μ infinite, $|A|, |B| \le \mu$. There is a closed pair $(A', B') \supset (A, B)$ with $|A'|, |B'| \le \mu$.

Proof. Alternate ω times: $B_{k+1} = B_k \cup \bigcup_{x \in H_k} \operatorname{supp} \sigma(x)$ (H_k a Hamel basis of P_{A_k}, $|H_k| \le \mu$ by the **hereditary dimension bound**), and $A_{k+1} = A_k \cup \bigcup_{s \in B_{k+1}} \operatorname{supp} \pi(e_s) \cup \bigcup_{\gamma \in A_k, n} \operatorname{supp} p_n^\gamma$. Take unions. (a), (c) are clear. For (b): $\bigcup_k P_{A_k}$ is dense in $\ell_1(A')$; for $x \in P_{A'}$ take $x_j \to x$ from it, then $\sigma(x_j) \in \ell_1(B')$, so $\sigma(x) \in \ell_1(B') \cap \ell_1^0(S) = \ell_1^0(B')$. □

Subtlety: $\operatorname{supp} x$ need not sit in one A_k, so density, not a support argument, proves (b).

Lemma (unions)

The union of an increasing chain $((A_\delta, B_\delta))_{\delta<\lambda}$ of closed pairs is a closed pair, and

$$P_{A_\lambda} = \pi(\ell_1^0(B_\lambda)) = \bigcup_{\delta<\lambda} \pi(\ell_1^0(B_\delta)) = \bigcup_{\delta<\lambda} P_{A_\delta}.$$

Proof. (a), (c) are inherited. (b): $\bigcup_\delta P_{A_\delta}$ is dense in $\ell_1(A_\lambda)$ (each P_{A_δ} is dense in $\ell_1(A_\delta)$), and the same continuity argument gives $\sigma(x) \in \ell_1^0(B_\lambda)$ for $x \in P_{A_\lambda}$. The formula for P_{A_λ} follows from $P_A = \pi(\ell_1^0(B))$ for closed pairs. □

Continuity (ii) is **automatic** along chains of closed pairs, for any cofinality. This is exactly what the ladder space lacks: there the retraction σ does not exist, and the closed spans jump at every limit.

Lemma (block retract)

Let $(A, B) \subset (A', B')$ be closed pairs, $N = A' \setminus A$. Then $D = \{x|_N : x \in P_{A'}\}$ is a dense subspace of $\ell_1(N)$ and a retract of $\ell_1^0(B' \setminus B)$; in particular D is topologically projective.

Proof. Density: $P_{A'}$ is dense in $\ell_1(A')$ and restriction is a contractive surjection. Retraction:

$$\rho\colon D \to \ell_1^0(B' \setminus B), \qquad \rho(y) = \sigma(x)|_{S \setminus B} \quad (x \in P_{A'},\ x|_N = y).$$

- well defined: two lifts differ by an element of P_A, whose σ-image lies in $\ell_1^0(B)$;
- bounded, $\|\rho\| \le \|\sigma\|$: choose the lift x with $\|x\| \le \|y\| + \varepsilon$, using density of P_A in $\ell_1(A)$;
- left inverse: $r\pi\rho(y) = r\big(\pi\sigma(x) - \pi(\sigma(x)|_B)\big) = r(x) - 0 = y$, since $\pi(\sigma(x)|_B) \in P_A \subset \ell_1(A)$ vanishes on N.

□So each block D_δ is again a **dense topologically projective subspace of a smaller** $\ell_1(N_\delta)$: the induction hypothesis applies.

Theorem

Let P be a dense topologically projective subspace of $\ell_1(\Gamma)$. For every $\theta > 0$ there is an automorphism T of $\ell_1(\Gamma)$ with $T(P) = \ell_1^0(\Gamma)$, $\|T\| \le 1 + \theta$, $\|T^{-1}\| \le 1$. In particular $d_{BM}(P, \ell_1^0(\Gamma)) = 1$ and P is extremally projective.

Proof: induction on $\kappa = |\Gamma|$.

- κ finite: $P = \ell_1(\Gamma)$. $\kappa = \aleph_0$: $\dim P \le \aleph_0$ by the dimension bound; countable back-and-forth.
- κ uncountable, $\lambda = \operatorname{cf}\kappa$. Write $\Gamma = \bigcup_{\delta<\lambda} \Gamma^\delta$ with $|\Gamma^\delta| < \kappa$ (singletons if κ is regular; initial segments of lengths $\kappa_\delta \nearrow \kappa$ if singular).
- Build closed pairs (A_δ, B_δ), $\delta \le \lambda$: $A_0 = B_0 = \varnothing$; successor: close off $(A_\delta \cup \Gamma^\delta, B_\delta)$, size $< \kappa$; limits: unions, size $< \kappa$ because fewer than $\operatorname{cf}\kappa$ sets of size $< \kappa$. $A_\lambda = \Gamma$.
- Hypotheses of the transfinite theorem for $\theta' = \theta/8$: (i) closed pairs are dense; (ii) unions lemma; (iii) block retract lemma + induction hypothesis on $|N_\delta| < \kappa$.
- $(1+\theta')^2(1-\theta')^{-1} \le 1 + \theta$; rescale. □

The blocks N_δ have size $< \kappa$, not necessarily countable: this is why the transfinite theorem allows arbitrary blocks.

The characterization in all dimensions

Theorem

For a normed space P the following are equivalent:

(i) *P is extremally projective;*

(ii) *P is topologically projective and $\overline{P} \cong \ell_1(\Gamma)$ isometrically;*

(iii) *$d_{BM}(P, \ell_1^0(\Gamma)) = 1$ for some set Γ;*

(iv) *for every $\varepsilon > 0$ there is Λ with $d_{BM}(P, \ell_1^0(\Lambda)) < 1 + \varepsilon$.*

In (iii), $|\Gamma| = \dim P$ for infinite dimensional P, and the distance is realized by bijections $T\colon \overline{P} \to \ell_1(\Gamma)$ with $T(P) = \ell_1^0(\Gamma)$ and $\|T\|\|T^{-1}\|$ arbitrarily close to 1.

Proof. (i)⇒(ii): completion theorem. (ii)⇒(iii): $J(P)$ is a dense topologically projective subspace of $\ell_1(\Gamma)$; general theorem. (iv)⇒(i): criterion. □

Grønbæk's conjecture

Every extremally projective normed space is **almost isometric** to $\ell_1^0(\Gamma)$, $|\Gamma| = \dim P$. The conjecture holds in the almost isometric form and fails in the isometric form.

Helemskii's question on dense subspaces

Which dense subspaces $P \subset \ell_1(\Lambda)$ are extremally projective?

Answer

P extremally projective $\iff$ P topologically projective $\iff$ P admits a filtration with (i) dense pieces, (ii) continuity, (iii) solvable blocks.

- Λ countable: $\iff \dim P \leq \aleph_0$.
- Λ uncountable: a genuine restriction beyond dimension. The ladder space is dense, has countable dimensional separable subspaces, satisfies (i) and (iii), fails (ii), and is not topologically projective.
- Topological and extremal projectivity coincide for dense subspaces of $\ell_1(\Gamma)$. This is a feature of the *exact* norm: if E is $\ell_1(\Gamma)$ with an equivalent, not almost isometric, norm, then $J(\ell_1^0(\Gamma)) \subset E$ is topologically but not extremally projective (its completion is E).

Fodor's lemma is used only in the negative direction. The positive results use only closing-off and unions of closed pairs.

Plan

1. Lifting problems
2. Basic theory
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Where each ingredient was consumed

ingredient	used for	where
universal property of $\ell_1^0(\Lambda)$	free spaces are extremally projective; canonical morphism π_P	§2
retraction σ onto a free space	supports are finite: dimension bound, Fodor obstruction, closed pairs	§3, §6
Grothendieck (via duality of $\overline{P}$)	completion is $\ell_1(\Gamma)$	§4
exact ℓ_1-norm: disjoint blocks add	$\|R\| \leq \delta$ in the one-vector lemma; $\|W_\delta - 1\| \leq \theta$ transfinitely	§4, §6
Hahn–Banach, Neumann series in $\overline{P}$	hitting lemma; invertibility	§4
density of P_δ in $\ell_1(\Gamma_\delta)$	corrections z_n at successor steps	§6
extreme points of B_{ℓ_1}	no space strictly between $\ell_1^0(S)$ and $\ell_1(S)$ is free	§5
Fodor's lemma	the ladder space is not projective	§6

Comparison with Grønbæk's proof

Grønbæk: topologically projective $\Rightarrow$ isomorphic to $\ell_1^0(\Lambda)$, in two steps.

1. Transfinite construction: every subspace $Y \subset \ell_1^0(\Lambda)$ contains a complemented copy of $\ell_1^0(\Lambda')$, $|\Lambda'| = \operatorname{dens} Y$.
   Inspection: the copy is $(1+\varepsilon)$-isomorphic and $(1+\varepsilon)$-complemented.
2. Pełczyński's decomposition method.
   Here the constants are lost.

- decomposition method: $Y \oplus \ell_1^0(\Lambda) \cong \ell_1^0(\Lambda)$, then $Y \cong Y \oplus \ell_1^0(\Lambda) \cong \ell_1^0(\Lambda)$, with uncontrolled constants;
- back-and-forth: an automorphism of $\ell_1(\Gamma)$ of distortion $< 1+\varepsilon$ moving P exactly onto $\ell_1^0(\Gamma)$.

The price of keeping the constants is the continuity condition (ii), and the closed pairs show that it is always available for projective spaces.

Open problems

- An *intrinsic* description of the topologically projective dense subspaces of $\ell_1(\Lambda)$ for uncountable Λ, not referring to a retraction onto a free space. The ladder space shows that "all separable subspaces have countable dimension" is not enough; ω_1-continuity is necessary. Is some continuity condition sufficient?
- Dense subspaces of other Banach spaces: for $E \ncong \ell_1(\Gamma)$ almost isometrically there are topologically but not extremally projective dense subspaces. Which E admit extremally projective dense subspaces at all? (Only those with $E \cong \ell_1(\Gamma)$ isometrically.)
- Extremal projectivity over non-zero algebras: Helemskii's original setting of sequence algebras. Is there an analogue of the almost isometric characterization, and of the counterexample?
- The role of set theory: Fodor's lemma entered through ω_1. Do larger cardinals bring new obstructions, or is the closed-pair argument the whole story?

The real question

In $\mathbf{Nor}$ the free objects are tiny and the projective objects are almost free. What replaces the Banach space toolbox when every complete subspace is finite dimensional?

Summary

- **Extremally projective** $= C$-topologically projective for every $C > 1$, in the category of normed spaces.
- **Counterexample**: $\ell_1^0(\mathbb{N}) + \mathbb{C}v$ is extremally projective and not free. The isometric conjecture of Helemskii and Grønbæk fails.
- **Characterization**: extremally projective $\iff$ topologically projective with completion $\ell_1(\Gamma) \iff d_{BM}(P, \ell_1^0(\Gamma)) = 1$. The conjecture holds almost isometrically.
- **Countable dimension**: a property of the completion alone.
- **Uncountable dimension**: not a property of the completion, not a property of the separable subspaces; Fodor's lemma and the ladder space.
- **Method**: back-and-forth, countable and transfinite, with perturbations on disjoint blocks.

Thank you!

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Backup: no 1-topologically projective spaces

Let $X = c_{00}$ with the sup-norm, $s = \sum_n 2^{-n}(1-2^{-n})$ and

$$f(x) = s^{-1} \sum_{n \geq 1} 2^{-n}(1-2^{-n})\, x_n, \qquad \|f\| \leq 1.$$

- f is coisometric: for $|z| < 1$ pick N with $|z| < s^{-1} \sum_{n \leq N} 2^{-n}(1-2^{-n})$ and $x = t \operatorname{sgn}(z) \sum_{n \leq N} e_n$ with suitable $t \in [0,1)$; then $f(x) = z$, $\|x\| < 1$.
- f does not attain its norm: $|f(x)| = 1$, $\|x\| \leq 1$ would force $x_n = \operatorname{sgn} f(x)$ for all n.

Let $P \neq 0$ be 1-topologically projective, $p \in P$, $\|p\| = 1$, and $g \in P^*$ with $\|g\| = g(p) = 1$ (Hahn–Banach). Lift g along f: $\psi\colon P \to X$, $f\psi = g$, $\|\psi\| \leq 1$. Then $x = \psi(p) \in B_X$ and $f(x) = g(p) = 1$. Contradiction. □

The same functional on ℓ_1, $x \mapsto \sum (1-2^{-n}) x_n$, shows that with the open-ball convention 1-topological projectivity is empty in **Ban** too. With closed balls (strictly coisometric operators) one recovers the metric theory; for extremal projectivity the choice is immaterial.

Backup: the product lemma

$\theta_m = \|F_m - 1_X\| < 1$, $\sum \theta_m < \infty$, $A_j = F_j \cdots F_1$, $M = \prod(1+\theta_m)$, $M' = \prod (1-\theta_m)^{-1}$.

- $\|A_j\| \le M$ and, by the Neumann series, $\|F_m^{-1}\| \le (1-\theta_m)^{-1}$, so $\|A_j^{-1}\| \le M'$.
- $\|A_j - A_{j-1}\| = \|(F_j - 1)A_{j-1}\| \le \theta_j M$: Cauchy, $A_j \to T$, $\|T\| \le M$.
- $\|A_j^{-1} - A_k^{-1}\| = \|A_j^{-1}(A_k - A_j)A_k^{-1}\| \le {M'}^2 \|A_k - A_j\|$: Cauchy, $A_j^{-1} \to B$, $\|B\| \le M'$.
- $A_j A_j^{-1} = A_j^{-1} A_j = 1$ in the limit: $TB = BT = 1$. □

In the countable back-and-forth the θ_m are chosen in advance with $\prod(1+\theta_m)/(1-\theta_m) \le 1+\varepsilon$; every step then has a fixed budget θ_m for its perturbation, and the budget is respected by the one-vector lemma ($\|R\| \le \delta$) and the hitting lemma ($\|S - 1\| \le 2C\|z - e_n\|$, with z chosen close enough to e_n by density).

Backup: the ladder space, details

Intersection of ladders. For limits $\alpha < \beta$: $\operatorname{supp} v_\alpha \subset \alpha$ and only finitely many β_n are $< \alpha$ because $\beta_n \nearrow \beta > \alpha$.

$P_L \cap \ell_1(\delta)$**.** Let $x = y + \sum_{\alpha \in F} c_\alpha v_\alpha \in \ell_1(\delta)$, $y \in \ell_1^0(\omega_1)$, $c_\alpha \neq 0$. If $\alpha_0 = \max F > \delta$, then $\operatorname{supp} v_{\alpha_0} \cap [\delta, \alpha_0)$ is infinite while $\operatorname{supp} y$ and $\operatorname{supp} v_\alpha \cap \operatorname{supp} v_{\alpha_0}$ $(\alpha \neq \alpha_0)$ are finite, so x has infinitely many non-zero coordinates in $[\delta, \alpha_0)$. Hence $F \subset [0, \delta]$ and $\operatorname{supp} y \subset \delta$:

$$P_L \cap \ell_1(\delta) = \ell_1^0(\delta) + \operatorname{span}\{v_\alpha : \alpha \leq \delta\},$$

countable dimensional and dense in $\ell_1(\delta)$, hence extremally projective.

Linear independence. In $\sum_F c_\alpha v_\alpha = 0$ with $\alpha_0 = \max F$, the coordinate at $\alpha_{0,n}$ equals $2^{-n} c_{\alpha_0}$ for all large n.

Separable subspaces. Their supports lie in a countable set, hence in some $P_L \cap \ell_1(\delta)$.